

Combining Markov Degradation Models and Data-Driven Diagnostics for Hidden-State Inference in Condition-Based Maintenance

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Abstract

Condition-based maintenance requires accurate information about the underlying degradation state. However, diagnostic outputs are uncertain and may vary across fault modes. This study investigates whether hidden-state estimation can be improved by combining diagnostic uncertainty with temporal degradation information. It develops a Markov degradation model for ball bearings, with separate early and advanced degradation states for outer-race, inner-race, and ball-defect faults. Transition probabilities are estimated from right-censored endpoint data. Diagnostic uncertainty is represented using the confusion-matrix observation model of a 1D convolutional neural network. Together, these sources form a Markov–diagnostic state-fusion procedure that uses Markov predictions and diagnostic likelihood correction to infer the hidden state.

The state-fusion procedure is evaluated in a simulated condition-based maintenance setting. Compared with the diagnostic baseline, the corrected-belief estimate increases state-estimation accuracy from 0.628 to 0.790. The largest gain is in the advanced degradation states, which are difficult for the diagnostic model to identify. A secondary maintenance evaluation shows that using the corrected operating belief in a threshold-based replacement policy reduces the cost rate by 6.6%, at the cost of more corrective replacements. Furthermore, the results also show that the quality of the estimated transition matrix is important. Weakly informed degradation branches can introduce model-based bias. Overall, the study demonstrates that the state-fusion procedure can improve hidden-state inference and support maintenance decisions.

Keywords: Condition-based maintenance, Belief updating, Diagnostic uncertainty, Markov degradation model, Ball bearings

1 Introduction

Condition-based Maintenance (CBM) uses condition-monitoring data to decide when a component should be replaced. This decision usually depends on the degradation state of the component. However, during operation, that state is not directly observed. Instead, it must be inferred from imperfect condition-monitoring information and from assumptions about how degradation evolves over time.

Two modelling perspectives address this problem from different directions. First, data-driven diagnostic models provide evidence about the current condition, but their outputs can be uncertain, noisy, and class-dependent. Second, Markov degradation models describe how hidden degradation states evolve between observations. However, their transition probabilities are not known and must be estimated from reliability data. Therefore, neither the diagnostic model nor the Markov model should be treated as perfect. The central modelling problem is how to combine these two sources of information so that they supplement each other in hidden-state inference.

Foundational work on partially observable Markov decision processes (POMDPs) established the representation of the physical state as hidden and the use of belief states to support decisions under imperfect information (Lovejoy, 1991; Smallwood & Sondik, 1973). Furthermore, indirect observations require inference about an underlying state process (Si et al., 2011). In data-driven diagnosis, convolutional neural networks (CNNs) and related models are increasingly used for fault classification, but their predictions remain uncertain and class-dependent (Nemani et al., 2023). These perspectives complement each other: diagnostic models provide evidence about the current condition, while Markov degradation models impose temporal consistency on the hidden state sequence.

The primary research question is whether hidden-state inference in a multi-mode, multi-severity degradation process can be improved by combining an estimated Markov degradation model with imperfect diagnostic information. A secondary question is whether this also improves threshold-based maintenance decisions under the tested cost structure.

The contribution of this study is a Markov–

diagnostic fusion procedure for hidden-state inference when both the diagnostic model and the Markov degradation model are imperfect, also referred to as the fusion procedure. The fusion procedure combines an estimated multi-mode Markov degradation model with class-dependent diagnostic likelihoods derived from a confusion-matrix observation model. The study further evaluates whether the resulting corrected operating belief improves both hidden-state estimation and threshold-based maintenance decisions in a multi-mode, two-severity bearing degradation setting.

The fusion procedure is developed for systems that follow a monotone, multi-mode, two-severity Markov degradation process. The true transition matrix is not assumed to be available and is estimated from right-censored endpoint data. The diagnostic model is represented by a confusion-matrix observation model, which captures class-dependent diagnostic uncertainty. At each observation time, the fusion procedure predicts the next operating-state distribution using the estimated transition matrix and then fuses this predicted operating prior with diagnostic evidence from the confusion-matrix observation model. The study then evaluates how the corrected operating belief affects both state-estimation accuracy and threshold-based maintenance decisions.

The numerical case study uses rolling-element bearings. For bearings, vibration-based monitoring is widely used because localized defects in the outer race, inner race, or rolling elements can produce measurable vibration signatures (Randall & Antoni, 2011; Tandon & Choudhury, 1999). These signatures are uncertain: early faults may be weak, and noise or normal operating vibrations can make a single diagnostic output unreliable (Randall & Antoni, 2011; Taghiyarrenani & Berenji, 2022; Tandon & Choudhury, 1999; Zheng et al., 2024). The case study builds on the diagnostic framework of Lee and Fecarotti (2026), which models bearing-condition classification under diagnostic uncertainty using one severity level for each fault mode. This study reuses their 1D-CNN diagnostic architecture, but retrains it for a seven-state operating space with one healthy state and separate early and advanced degradation states for each fault mode. This case study is used to evaluate whether the proposed fusion procedure improves state estimation and replacement timing.

The evaluation has two parts. First, this study tests whether the corrected operating belief improves state-estimation accuracy relative to the diagnostic baseline. Second, the corrected operating belief is used in threshold-based replacement policies and compared with a diagnostic-evidence threshold policy. This tests whether improved state inference translates into better replacement timing.

The remainder of this paper is structured as follows. Section 2 defines the general degradation state space, diagnostic information, right-censored transition data, and inference objectives. Section 3 presents the transition-matrix estimation procedure and the Markov–diagnostic fusion procedure. Section 4 intro-

duces the ball-bearing numerical case study and evaluation design. Section 5 reports the results. Section 6 discusses the implications and limitations of the findings, and Section 7 concludes the paper.

2 Problem Setting

2.1 System and degradation states

Consider a single component observed at discrete time periods $t = 0, 1, 2, \dots$, corresponding to one observation interval in degradation. At each time step, the component occupies a true degradation state S_t that is not directly observed.

Let M denote the set of possible fault modes. The degradation process is represented by a two-severity state space.

$$S = \{H, F\} \cup \{m_1, m_2 : m \in M\}.$$

Here, H denotes the healthy state and F denotes failure. $m \in M$ denotes a fault mode. The subscript indicates degradation severity: 1 denotes early degradation, and 2 denotes advanced degradation.

The model follows a monotone degradation structure. A component starts in H and enters at most one fault branch. Within this branch, it progresses from severity 1 to severity 2, and then reaches failure:

$$H \rightarrow m_1 \rightarrow m_2 \rightarrow F.$$

Transitions between fault modes are not allowed, and recovery transitions are excluded. Failure is assumed to be self-announcing.

Let $P = (p_{ij})$ denote the transition probability matrix, where

$$p_{ij} = \mathbb{P}(S_{t+1} = j \mid S_t = i). \quad (2.1.1)$$

All structurally inadmissible transitions are fixed at zero. Each row of P sums to one, and F is absorbing.

For inference, the relevant state space is the set of transient states. Failure is excluded because it is self-announcing.

$$T = S \setminus \{F\}.$$

2.2 Diagnostic information

At each observation time, condition-monitoring information is obtained. This information is processed by an imperfect diagnostic model. Its error behaviour is represented by a class-conditional observation model over the transient states T .

The diagnostic model produces a hard diagnostic label $\hat{s}_t \in T$. Let G denote the row-normalized confusion matrix of the diagnostic model, used as the confusion-matrix observation model, with entries

$$G_{ij} = \mathbb{P}(\hat{s}_t = j \mid S_t = i), \quad i, j \in T. \quad (2.2.1)$$

Here, $S_t = i$ is the true operating state and $\hat{s}_t = j$ is the hard diagnostic label. Each row of G describes

the distribution of the diagnostic predictions conditional on the true state. Therefore, G is used as the confusion-matrix observation model that captures diagnostic uncertainty.

During simulation evaluation, the true state S_t is known from the generated degradation trajectory. If $S_t = i$, the diagnostic label is generated as

$$\hat{s}_t \sim \text{Categorical}(G_{i,:}).$$

This is a hard diagnostic label with the same class-conditional error probability as the diagnostic model.

For the maintenance evaluation based on diagnostic evidence, $\hat{s}_t$ is converted into a diagnostic-evidence vector using the corresponding column of the confusion matrix. For an observed label $\hat{s}_t = j$, define

$$\tilde{g}_t(s) = \frac{G_{sj}}{\sum_{u \in T} G_{uj}}, \quad s \in T. \quad (2.2.2)$$

$\tilde{g}_t(s)$ is a normalized diagnostic likelihood. It uses the current diagnostic label and the empirical confusion matrix.

For the fusion procedure, the same diagnostic label is used through the unnormalized likelihood vector $G_{sj} = \mathbb{P}(\hat{s}_t = j \mid S_t = s)$. This describes how compatible each possible hidden state s is with the observed diagnostic label $\hat{s}_t = j$. Thus, diagnostic labels are simulated from a row of G , while $\tilde{g}_t(s)$ and the fusion procedure use the column of G corresponding to the observed diagnostic label.

2.3 Right-censored transition data

The fusion procedure requires a transition matrix P . This matrix is not assumed to be known and must be estimated from degradation data.

The data considered in this study are right-censored. Each observed unit has either failed or is censored. For a failed unit, the failure time τ_F and the fault mode $m \in M$ are recorded. A censored unit has not failed at the censoring time τ_C , the information it provides is the endpoint state $S_C = s$ at τ_C .

In the simulation, the censored states are available from generated degradation trajectories. The estimation problem is to obtain a transition matrix P that is consistent with the monotone degradation structure and with the observed failed and censored data. The resulting matrix is then used as the transition model in the fusion procedure.

2.4 Inference and decision objectives

The primary objective is to estimate the hidden degradation state during operation. Let $b_t(s)$ denote the corrected operating belief assigned to state $s \in T$ after diagnostic outputs up to time t have been observed, with

$$b_t(s) \geq 0, \quad \sum_{s \in T} b_t(s) = 1.$$

$b_t(s)$ is a probability distribution over the transient degradation states, conditional on the component still

operating. It uses the sequence of diagnostic outputs together with the Markov degradation model.

As a secondary objective, the corrected operating belief is used to support maintenance decisions, such as replacing a bearing. This decision objective is treated separately from the inference objective because improved state estimation does not necessarily imply improved maintenance performance. The final maintenance outcome also depends on how the corrected operating belief is converted into a replacement decision.

3 Methodology

3.1 Transition-matrix estimation

The fusion procedure requires a transition matrix P , estimated from the right-censored data described in Section 2.3.

The full degradation path is unobserved. The estimator computes the expected number of transitions of each admissible type, conditional on the observed failure time or censored endpoint state. These expected counts are normalized row-wise to update the transition probabilities.

Let Q_m denote the transient submatrix of P restricted to the states $\{H, m_1, m_2\}$. Let f_m denote the vector of transition probabilities from these states to F . For a unit that fails in mode m at time τ_F , the likelihood contribution is

$$L_F(m, \tau_F; P) = e_H^\top Q_m^{\tau_F-1} f_m, \quad (3.1.1)$$

where e_H is a unit vector corresponding to the healthy state. This likelihood sums over all unobserved internal paths that remain in branch m and reach failure exactly at time τ_F .

For censored units, the endpoint state is directly observed. Let

$$Q = P[T, T]$$

denote the transient block of the full transition matrix. For a unit censored at time τ_C in state $s \in T$, the likelihood contribution is

$$L_C(s, \tau_C; P) = e_H^\top Q^{\tau_C} e_s, \quad (3.1.2)$$

where e_s is the unit vector corresponding to endpoint state s . This term gives the probability that a unit starting in H occupies state s at the censoring time without having reached failure.

Because all units in the same cells have identical likelihood contributions, estimation can be based on the grouped observed-data log-likelihood. So, the data are grouped by failed mode-time cells and censored state-time cells. Let $\mathcal{R}_F$ denote the set of failed units and let $\mathcal{R}_C$ denote the set of censored units. For unit r , let $S_t^{(r)}$ denote the true degradation state at time t . For $r \in \mathcal{R}_F$, let $M_r \in M$ denote the observed failure mode and let $\tau_{F,r}$ denote the observed failure time. For $r \in \mathcal{R}_C$, let $\tau_{C,r}$ denote the censoring time and let $S_{\tau_{C,r}}^{(r)} \in T$ denote the observed endpoint state at censoring.

Let

$$\mathcal{T}_F = \{\tau_{F,r} : r \in \mathcal{R}_F\} \quad (3.1.3)$$

denote the set of observed failure-time cells, and let

$$\mathcal{T}_C = \{\tau_{C,r} : r \in \mathcal{R}_C\} \quad (3.1.4)$$

denote the set of observed censoring-time cells. For $m \in M$ and $t \in \mathcal{T}_F$, define

$$n_F(m, t) = \#\{r \in \mathcal{R}_F : M_r = m, \tau_{F,r} = t\} \quad (3.1.5)$$

as the number of failed units observed in mode-time cell (m, t) . For $s \in T$ and $c \in \mathcal{T}_C$, define

$$n_C(s, c) = \#\{r \in \mathcal{R}_C : S_{\tau_{C,r}}^{(r)} = s, \tau_{C,r} = c\} \quad (3.1.6)$$

as the number of censored units observed in state-time cell (s, c) .

The grouped observed-data log-likelihood is given by,

$$\begin{aligned} \ell_{\text{obs}}(P) = & \sum_{m \in M} \sum_{t \in \mathcal{T}_F} n_F(m, t) \log L_F(m, t; P) \\ & + \sum_{s \in T} \sum_{c \in \mathcal{T}_C} n_C(s, c) \log L_C(s, c; P). \end{aligned} \quad (3.1.7)$$

The transition matrix is estimated by maximizing $\ell_{\text{obs}}(P)$ subject to the monotone transition structure. This right-censored EM estimator produces the estimated transition matrix $\hat{P}$. Estimation is performed using an expectation-maximization (EM) algorithm. In the E-step, the expected number of transitions $i \rightarrow j$ is computed conditional on the observed failed and censored data and the current transition matrix $P^{(k)}$.

Let $\bar{N}_{ij}^{(k)}$ denote the expected number of transitions from state i to state j , conditional on the observed failed and censored data and the transition matrix $P^{(k)}$. The M-step updates each admissible row by normalizing these expected transition counts:

$$p_{ij}^{(k+1)} = \frac{\bar{N}_{ij}^{(k)}}{\sum_{\ell \in A(i)} \bar{N}_{i\ell}^{(k)}}, \quad j \in A(i). \quad (3.1.8)$$

Here, $A(i)$ is the set of admissible successor states from i . Structural zeros are preserved by excluding inadmissible transitions, and failure remains absorbing with $P_{FF} = 1$. The full E-step recursions used to compute $\bar{N}_{ij}^{(k)}$ are given in Appendix A. Implementation details are reported in Appendix B. This estimator is referred to as the EM transition-matrix estimator in the remainder of the paper.

This estimation provides the estimated transition matrix $\hat{P}$ used in the fusion procedure. The main contribution is the combination of this estimated transition matrix with the confusion-matrix observation model.

3.2 Markov-diagnostic fusion procedure

The purpose of the fusion procedure is to combine the current diagnostic label with the degradation history

implied by the Markov degradation model. The estimated transition matrix is denoted by $\hat{P}$ with entries $\hat{p}_{ij}$. The operating belief at time t is denoted over the state space T .

After installation or replacement, the component is assumed to be healthy. Therefore, the initial operating belief is

$$b_0(H) = 1, \quad b_0(s) = 0 \quad \text{for } s \in T, s \neq H.$$

At each observation time, the update first performs a Markov prediction step. Given the previous belief b_{t-1} , the unnormalized predicted operating prior is

$$\bar{a}_t(s) = \sum_{r \in T} b_{t-1}(r) \hat{p}_{rs}, \quad s \in T. \quad (3.2.1)$$

This prediction uses the estimated transition matrix to propagate the previous belief one step forward. Since probability mass may move from a transient state to failure, this prediction is normalized over the operating state space:

$$a_t(s) = \frac{\bar{a}_t(s)}{\sum_{u \in T} \bar{a}_t(u)}, \quad s \in T. \quad (3.2.2)$$

The vector a_t is the predicted operating prior before the current diagnostic label is incorporated. This quantity is referred to as the prior for short. It is always the prior over transient operating states, conditional on the component still operating.

The diagnostic correction step then uses the hard diagnostic label observed at time t . If the diagnostic model produces label $\hat{s}_t = j$, the likelihood of this label under each possible hidden state is obtained from the corresponding column of the confusion matrix:

$$\ell_t(s) = \mathbb{P}(\hat{s}_t = j \mid S_t = s) = G_{sj}, \quad s \in T. \quad (3.2.3)$$

A likelihood floor $\epsilon = 10^{-4}$ is applied to avoid treating empirical zeros in the confusion matrix as impossible diagnostic events. Each likelihood is replaced by $\max\{G_{sj}, \epsilon\}$ before normalization.

The corrected operating belief is then obtained by combining the prior $a_t(s)$ with the diagnostic likelihood $\ell_t(s)$. If both the estimated transition matrix and the confusion-matrix observation model were treated as correctly specified, this would give the standard Bayesian correction

$$b_t^{\text{Bayes}}(s) = \frac{a_t(s) \ell_t(s)}{\sum_{u \in T} a_t(u) \ell_t(u)}, \quad s \in T. \quad (3.2.4)$$

In this version, $a_t(s)$ as the prior and $\ell_t(s)$ as the observation evidence. However, this assumes that both models are correctly specified, which is not the case in this study. Therefore, this study uses a tempered correction:

$$b_t(s) = \frac{a_t(s)^\alpha \ell_t(s)^\beta}{\sum_{u \in T} a_t(u)^\alpha \ell_t(u)^\beta}, \quad s \in T. \quad (3.2.5)$$

For $\alpha = \beta = 1$, Eq. (3.2.5) reduces to the standard

Bayesian posterior operating belief in Eq. (3.2.4). For general values of α and β , b_t is treated as a corrected operating belief rather than as an exact posterior under the original hidden Markov model. The powered-product form is a log-linear fusion rule, it combines nonnegative information sources by multiplying powered inputs and normalizing the result. This is closely related to log-linear pooling, or weighted geometric pooling, as discussed in the probability-density fusion literature (Koliander et al., 2022).

This formulation preserves the prior-likelihood structure of Bayesian filtering. It also allows the influence of the Markov prediction and the diagnostic likelihood to be calibrated. The parameter α is the prior exponent and controls the influence of the predicted operating prior, while β is the diagnostic-likelihood exponent and controls the influence of the diagnostic likelihood. Values $\alpha < 1$ or $\beta < 1$ flatten the corresponding source and reduce its influence; values greater than one sharpen it.

The values of α and β are selected on the training trajectories by grid search, with the selected pair fixed before held-out evaluation, as reported in Appendix B.

If failure is observed, the operating inference problem ends and the belief is assigned to the failed state. Otherwise, the corrected operating belief b_t is carried forward to the next observation time.

For the diagnostic baseline, the hard diagnostic label $\hat{s}_t$ is used as the diagnostic estimate. For the diagnostic-evidence threshold policy used in the maintenance evaluation, the diagnostic-evidence vector $\hat{g}_t$ is used. For the corrected-belief estimate, the inferred state is obtained via

$$\text{corrected-belief estimate} = \arg \max_{s \in T} b_t(s) \quad (3.2.6)$$

and compared with the true simulated state. This comparison tests whether the fusion procedure improves state-estimation accuracy relative to the diagnostic baseline.

The numerical instantiation and evaluation design are described separately in Section 4.

4 Case Study

The numerical case study instantiates the fusion procedure outlined in Section 3.2 for a rolling-element bearing. It also defines the evaluation design used to test state estimation and maintenance performance.

4.1 Bearing state space and severity labelling

For the bearing case study, the fault modes are $M = \{O, I, B\}$, where O , I , and B denote outer-race, inner-race, and ball-defect fault modes. The resulting degradation state space is

$$S = \{H, O_1, O_2, I_1, I_2, B_1, B_2, F\}.$$

The bearing case study follows the monotone degradation structure outlined in Section 2.1. The allowed

transitions are shown in Figure 1.

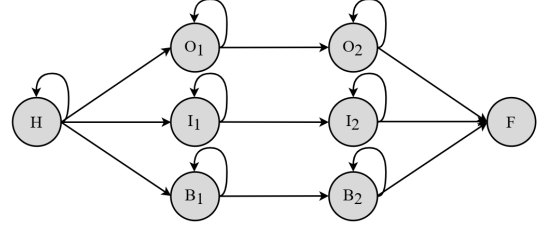


Figure 1: Severity-expanded Markov degradation model with two degradation levels for each fault mode.

The corresponding transition matrix P has the structure

$$P = \begin{bmatrix} p_{H,H} & p_{H,O_1} & 0 & p_{H,I_1} & 0 & p_{H,B_1} & 0 & 0 \\ 0 & p_{O_1,O_1} & p_{O_1,O_2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & p_{O_2,O_2} & 0 & 0 & 0 & 0 & p_{O_2,F} \\ 0 & 0 & 0 & p_{I_1,I_1} & p_{I_1,I_2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & p_{I_2,I_2} & 0 & 0 & p_{I_2,F} \\ 0 & 0 & 0 & 0 & 0 & p_{B_1,B_1} & p_{B_1,B_2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & p_{B_2,B_2} & p_{B_2,F} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

with row and column order $(H, O_1, O_2, I_1, I_2, B_1, B_2, F)$. For inference during operation, the transient state space is $T = S \setminus \{F\}$.

The two-severity model requires each fault observation to be assigned to either an early or advanced degradation state. Severity is operationalized using physical fault length Δl , according to the fault-evolution interpretation in Zheng et al. (2024). They distinguish three evolution stages. However, to keep the state space small and manageable, the first two stages, fault generation and slow evolution, are aggregated into severity level 1. The accelerated-evolution stage is assigned to severity 2. The following thresholds are used for each fault mode M :

$$\begin{aligned} M_1 : 0 \leq \Delta l < 0.5 \text{ mm}, \\ M_2 : 0.5 \leq \Delta l \leq 0.8 \text{ mm}. \end{aligned} \quad (4.1.1)$$

These thresholds are a modelling choice.

4.2 Diagnostic model for the case study

Following the diagnostic framework of Lee and Fecarotti (2026), a 1D-CNN diagnostic model is trained to classify the bearing condition from vibration data. In this study, the 1D-CNN is trained directly on the state space T . The model is trained as a seven-class classifier, assigning each operating condition to one of the transient degradation states.

The training data consist of 60 cases for each fault state and 60 cases for the healthy state H . For the six fault states O_1, O_2, I_1, I_2, B_1 , and B_2 , the fault lengths are sampled uniformly within the limits defined in Eq. (4.1.1). The validation set consists of 12 cases per operating state and is used during model development. The empirical confusion matrix is generated from a held-out test set of 15 cases per state.

To avoid repeatedly evaluating the CNN inside the simulation study, the trained 1D-CNN diagnostic model is represented by its empirical confusion matrix.

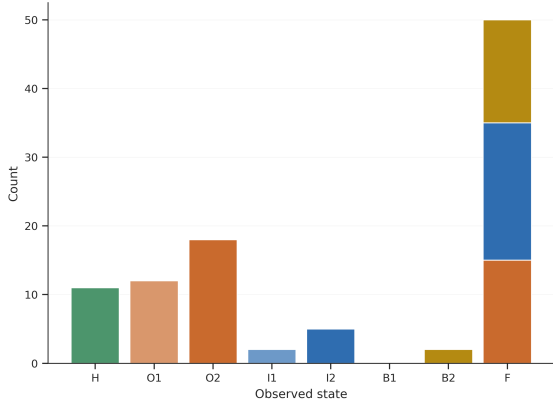


Figure 2: Counts for each censored and failed bearing.

This empirical matrix defines the confusion-matrix observation model G used as the simulation and inference proxy described in Section 2.2.

4.3 Evaluation design

The evaluation has two objectives. The primary objective is to test whether the fusion procedure improves state accuracy relative to the diagnostic baseline. The second objective is to test whether the fusion procedure improves maintenance decisions.

To evaluate the fusion procedure, bearing trajectories are generated from the Markov degradation model. The data-generating transition matrix is denoted by P_0 and is given by

$$P_0 = \begin{bmatrix} 0.90 & 0.05 & 0 & 0.03 & 0 & 0.02 & 0 & 0 \\ 0 & 0.90 & 0.10 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.90 & 0 & 0 & 0 & 0 & 0.10 \\ 0 & 0 & 0 & 0.80 & 0.20 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.80 & 0 & 0 & 0.20 \\ 0 & 0 & 0 & 0 & 0 & 0.65 & 0.35 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.55 & 0.45 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The rows and columns follow the order $(H, O_1, O_2, I_1, I_2, B_1, B_2, F)$.

Two disjoint sets of simulated bearing trajectories are used. First, 100 trajectories are generated and right-censored at $\tau_C = 20$. These trajectories are used to estimate $\hat{P}$. Of these 100 trajectories, 50 bearings fail before the censoring time and 50 bearings are censored. Their distribution is shown in Figure 2.

Second, 1000 additional trajectories are generated to evaluate the accuracy of the fusion procedure and its effect on maintenance policies. These trajectories are fully disjoint from the transition-estimation data. Of these trajectories, 700 are used to tune the prior exponent α , the diagnostic-likelihood exponent β in the tempered correction in Eq. (3.2.5), and the maintenance policy thresholds. The remaining 300 test trajectories are held out and used to evaluate the model. All simulated bearings start in H at $t = 0$. Because failures are treated as self-announcing, only pre-failure operating observations are included for state-accuracy estimation.

Along each trajectory, the true operating state S_t is recorded for evaluation. Given this state, a diag-

nostic label $\hat{s}_t$ is sampled from the confusion-matrix observation model G , as described in Section 2.2.

State-estimation accuracy is computed over all operating observations in the test trajectories:

$$\text{Accuracy} = \frac{\sum_r \sum_{t \in \mathcal{T}_r} \mathbf{1}\{\hat{S}_t^r = S_t^r\}}{\sum_r |\mathcal{T}_r|}, \quad (4.3.1)$$

where $\mathcal{T}_r$ is the set of pre-failure operating observation times for trajectory r . The main comparison is between the diagnostic estimate $\hat{s}_t$ and the corrected-belief estimate.

The secondary objective is the maintenance evaluation. This study proposes two condition-based replacement (CBR) policies. Similar to Lee and Fecarotti (2026), these are threshold-based. The first policy, CBR0, is the diagnostic-evidence threshold policy. It uses the current-observation diagnostic-evidence vector $\tilde{g}_t$ and replaces when the evidence assigned to an advanced degradation state exceeds a mode-specific threshold:

$$\tilde{g}_t(O_2) > \eta_O \quad \text{or} \quad \tilde{g}_t(I_2) > \eta_I \quad \text{or} \quad \tilde{g}_t(B_2) > \eta_B.$$

The second policy, CBR1, is the corrected-belief threshold policy. It uses the corrected operating belief b_t . Replacement is again triggered if the belief assigned to an advanced degradation state exceeds a mode-specific threshold:

$$b_t(O_2) > \eta_O \quad \text{or} \quad b_t(I_2) > \eta_I \quad \text{or} \quad b_t(B_2) > \eta_B.$$

The thresholds (η_O, η_I, η_B) are selected on the training trajectories and then fixed before the test evaluation.

Maintenance performance is measured by cost rate,

$$\text{Cost rate} = \frac{\text{total maintenance cost}}{\text{total operating time}}. \quad (4.3.2)$$

The cost evaluation is reported separately from the state-estimation evaluation because a more accurate state estimate does not necessarily imply a lower maintenance cost. The final cost also depends on how the corrected operating belief is converted into a replacement decision.

Further implementation details for the EM initialization and threshold grid are reported in Appendix B.

5 Results

5.1 Transition-matrix estimation

This subsection reports the numerical results for the transition-matrix estimation of the bearing case study. The algorithm terminated because the convergence criteria for both the log-likelihood and the transition matrix were satisfied. Appendix B.1 reports the full details.

Table 1 shows the expected source counts, expected exit counts, and fitted exit probabilities at convergence. The expected source count represents the total expected number of times that a state is occupied,

while the expected exit count represents the expected number of departures from that state. Their ratio gives the fitted one-step exit probability.

Table 1: Expected statistics at convergence for the right-censored EM estimator.

State	$\mathbb{E}[\text{source count}]$	$\mathbb{E}[\text{exit count}]$	$\hat{q}_s$
H	839.19	89	0.11
O_1	284.84	33	0.12
O_2	186.19	15	0.08
I_1	82.76	25	0.30
I_2	100.80	20	0.20
B_1	17.01	17	0.9997
B_2	76.22	15	0.20

Applying the EM algorithm from Section 3.1 to the case-study data gives the estimated transition matrix

$$\hat{P} = \begin{bmatrix} 0.89 & 0.06 & 0 & 0.03 & 0 & 0.02 & 0 & 0 \\ 0 & 0.88 & 0.12 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.92 & 0 & 0 & 0 & 0 & 0.08 \\ 0 & 0 & 0 & 0.70 & 0.30 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.80 & 0 & 0 & 0.20 \\ 0 & 0 & 0 & 0 & 0 & 0.0003 & 0.9997 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.80 & 0.20 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The entries are rounded and the rows and columns follow the order $(H, O_1, O_2, I_1, I_2, B_1, B_2, F)$.

The estimated matrix $\hat{P}$ shows that the healthy state is persistent. The outer-race branch degrades most slowly, while the inner-race branch progresses more quickly. Both branches are estimated reasonably accurately. The largest discrepancy occurs in the ball-defect branch. Specifically, $\hat{p}_{B_1, B_2} = 0.9997$, implying almost immediate progression from B_1 to B_2 . This is consistent with Figure 2 and the data-generating matrix P_0 . The ball-defect branch is the least frequently observed branch and progresses quickly, so the right-censored data contain little information about time spent in B_1 . Together, this indicates that the right-censored data provide useful information for estimating the degradation process, but do not fully resolve the internal severity split for the ball-defect branch.

The censored observations include the transient states occupied at the censoring time, which helps distinguish units that remain in early degradation from units that have already reached advanced degradation. At the same time, the fitted ball-defect branch shows that finite-sample ambiguity remains when endpoint information is limited for a particular branch.

This transition-estimation result is important for the subsequent inference evaluation because it shows both the usefulness and the limitations of the estimated transition matrix used in the fusion procedure.

5.2 State-estimation performance

This subsection evaluates whether the fusion procedure improves hidden-state estimation relative to the diagnostic baseline. The comparison is performed on the held-out evaluation split. Only pre-failure operating observations are included because failure is as-

sumed to be self-announcing and is excluded from the operating state-estimation problem. In total, the test set contains 300 bearings and 6,882 pre-failure observations. The prior exponent α and the diagnostic-likelihood exponent β were tuned on the training split as described in Section 3.2, resulting in $\alpha = 1.0$ and $\beta = 1.8$.

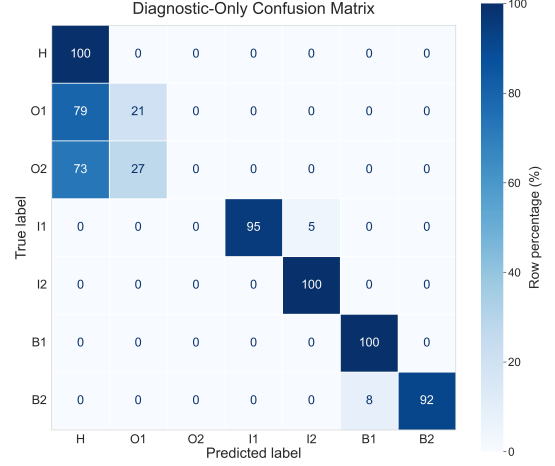


Figure 3: Row-normalized confusion matrix for the diagnostic estimate.

Figure 3 shows the confusion matrix for the diagnostic estimate. This matrix represents the diagnostic information available to the inference model. The diagnostic estimate has an observation-weighted accuracy of 0.628, calculated with Eq. (4.3.1). The balanced accuracy, corresponding to the average diagonal value of the row-normalized confusion matrix, is 0.725.

The diagnostic accuracy is strongly class-dependent. The healthy state is always identified correctly, and the inner-race and ball-defect states generally have high recall. However, the outer-race severity states are much more difficult. The recall for O_1 is 0.207, and for O_2 it is 0.000. These findings are consistent with Lee and Fecarotti (2026), where the outer-race fault mode also shows a high false-negative rate. Their model reported an accuracy of 0.85 on the four-state classification task (H , O , I , and B). The lower diagnostic-estimate accuracy (0.628) observed in this study is expected because the problem is more granular. Therefore, the diagnostic-baseline results indicate that the two-severity diagnostic task is more difficult than the one-severity fault-mode classification task.

After establishing the diagnostic baseline, the tempered correction in Eq. (3.2.5) is applied to combine the predicted operating prior with the diagnostic likelihood. The resulting corrected-belief estimates are shown in Figure 4.

Figure 4 shows the corresponding confusion matrix after applying the fusion procedure. The corrected-belief estimate increases the observation-weighted accuracy from 0.628 to 0.790, an absolute improvement of 0.162. Balanced accuracy also increases from 0.725 to 0.772.

Table 2 summarizes the state-estimation results. The largest improvements occur in the advanced

Table 2: State-estimation accuracy on held-out operating observations.

Estimator	Accuracy	Balanced accuracy	Severity-1 exact	Severity-2 exact
Diagnostic estimate	0.628	0.725	0.416	0.300
Corrected-belief estimate	0.790	0.772	0.483	0.808

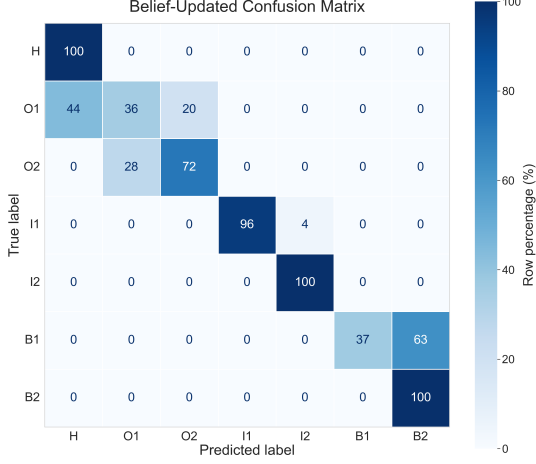


Figure 4: Row-normalized confusion matrix for the corrected-belief estimate.

degradation states. This mainly occurs because the diagnostic estimate never correctly identifies O_2 , whereas the corrected-belief estimate obtains a recall of 0.724 for O_2 . The fusion procedure therefore helps recover advanced degradation states that are poorly identified by the diagnostic baseline.

This improvement is not uniform across all states. For B_1 , recall decreases from 1.000 to 0.370 under the corrected-belief estimate. This is consistent with the transition-matrix estimation result, where the ball-defect branch contains limited information and the fitted model moves probability quickly from B_1 to B_2 . At the same time, the corrected-belief estimate improves B_2 recall from 0.916 to 1.000. This tradeoff is relevant for maintenance decisions, where identifying the correct advanced degradation state may be more important than exact early-state classification.

The values in Table 2 are point estimates computed on the held-out test trajectories. Appendix C reports a trajectory-level bootstrap analysis.

Overall, the held-out results show that the corrected-belief estimate improves overall state-estimation accuracy relative to the diagnostic estimate, with the largest gains occurring in the advanced degradation states.

These results answer the primary research question positively: combining the estimated Markov degradation model with diagnostic likelihoods improves hidden-state estimation relative to the diagnostic baseline.

5.3 Maintenance-cost performance

This subsection evaluates whether the improved state estimation translates into better maintenance deci-

sions in the bearing case study. First, the prior exponent α and diagnostic-likelihood exponent β are selected for state-estimation accuracy. Then, the policy thresholds are tuned for cost on the training split before evaluation on the held-out bearings.

The two proposed policies (CBR0 and CBR1) are compared in a scenario where corrective replacements are more expensive than preventive replacements. In this case, the preventive cost is $C_P = 10$ and the corrective cost is $C_C = 50$. Replacement is triggered when the evidence assigned to an advanced degradation state exceeds a threshold. For CBR0, this is done using $\tilde{g}_t$ from Eq. (2.2.2). Because $\tilde{g}_t$ depends only on the fixed confusion matrix, CBR0 is a memoryless diagnostic-evidence threshold policy. Once the confusion matrix G and the thresholds are fixed, each possible diagnostic label is mapped to a fixed replacement or continuation decision. For CBR1, the same threshold structure is applied to b_t . In contrast to CBR0, CBR1 can assign different replacement decisions to the same diagnostic label at different times.

The policy thresholds are selected on the training trajectories and then fixed for the held-out evaluation. For CBR0, the selected thresholds are

$$\eta_O = 0.5, \quad \eta_I = 0.9, \quad \eta_B = 0.9.$$

For CBR1, the selected thresholds are

$$\eta_O = 0.4, \quad \eta_I = 0.9, \quad \eta_B = 0.9.$$

Because CBR0 uses the normalized confusion-matrix column rather than a one-hot label, these thresholds can trigger replacement even when the observed hard diagnostic label is not itself an advanced state. For example, an observed label may still assign substantial diagnostic evidence to O_2 through $\tilde{g}_t(O_2)$. The lower outer-race thresholds reflect the weaker diagnostic performance for the outer-race branch observed in Section 5.2. The higher thresholds for the inner-race and ball-defect branches indicate that stronger evidence is required before preventive replacement is triggered for these modes.

Table 3 shows the maintenance-cost results. The diagnostic-evidence threshold policy obtains a cost rate of 0.730, while the corrected-belief threshold policy obtains a cost rate of 0.682. The absolute reduction is 0.048, which corresponds to a relative reduction of 6.6%.

CBR1 has a higher expected cost per bearing of 10.400, compared with 10.000 for CBR0. The improvement comes from a longer average operating cycle. The expected cycle length increases from 13.700 to 15.257 time steps. Therefore, using the corrected operating

Table 3: Held-out maintenance performance of diagnostic-evidence and corrected-belief threshold policies.

Policy	Cost rate	Cost/bearing	Cycle length	Preventive	Corrective
CBR0: diagnostic-evidence threshold policy	0.730	10.000	13.700	300	0
CBR1: corrected-belief threshold policy	0.682	10.400	15.257	297	3

belief b_t allows the simulated bearings to remain in operation longer before replacement.

The maintenance comparison is based on simulated held-out trajectories. Therefore, the uncertainty of the cost-rate estimates is evaluated using a trajectory-level bootstrap procedure. The results are shown in Appendix C.

The maintenance results show that the corrected-belief threshold policy obtains a lower held-out cost rate than the diagnostic-evidence threshold policy under the tested thresholds and cost parameters. This lower cost rate is achieved through longer operating cycles, but it is accompanied by three corrective replacements. The diagnostic-evidence threshold policy has no corrective replacements in the held-out test set.

The maintenance evaluation therefore suggests that improved inference can support better replacement timing, but the cost benefit depends on the threshold policy and cost structure used in the case study.

6 Discussion

The results show that temporal degradation information can improve hidden-state inference when diagnostic outputs are uncertain and class-dependent. The fusion procedure constrains diagnostic evidence using the degradation history implied by the Markov degradation model. The diagnostic model provides local evidence about the current condition, while the Markov degradation model provides temporal structure across observations. The corrected operating belief combines these two sources and uses more information than the diagnostic baseline.

6.1 Inference improvements

The main improvement occurs because the inference update introduces temporal consistency into the state-estimation problem. The diagnostic baseline treats each observation as a separate classification problem. Therefore, it cannot use the fact that degradation is monotone, that fault modes do not switch, or that advanced degradation states are more likely after a component has already shown earlier evidence of the same fault branch.

The fusion procedure uses this sequential structure. The Markov prediction step carries forward information from previous observations. Its benefits are most evident when the diagnostic model is uncertain and the temporal degradation structure provides additional information about which states are plausible. This is shown by the outer-race branch in the case study. The diagnostic model struggles to distinguish between outer-race degradation stages and the healthy state,

and the fusion procedure can use the degradation history to improve state inference. When the diagnostic model is already accurate, this benefit is smaller. Therefore, the value of the fusion procedure lies in resolving ambiguity in diagnostic outputs through the incorporation of temporal degradation information.

This also explains why the improvement should not be interpreted as a only an increase in classifier accuracy. The corrected-belief estimate is a sequential state estimator, whose performance depends on the interaction between diagnostic information, the estimated transition matrix, and the accumulated prior.

6.2 Model bias

The same mechanism that improves inference can also introduce bias. The Markov degradation model shapes which state sequences are plausible. If the estimated transition matrix is weakly informed, the fusion procedure can move probability mass too quickly or too slowly through a degradation branch.

The ball-defect branch from the case study illustrates this limitation. The fitted transition matrix implies almost immediate progression from B_1 to B_2 . As a result, the corrected-belief estimate improves recall for B_2 , but reduces recall for B_1 . This tradeoff shows that the fusion procedure is useful when the Markov degradation model captures the relevant state dynamics. However, it can become overconfident when the estimated transition matrix is weakly informed. In practical applications, the quality of the estimated transition matrix should be assessed before relying on the corrected belief. Branch imbalance, short dwell times, and limited censoring information can all affect the reliability of the estimated transition probabilities.

The tempered correction partly addresses this issue by allowing the influence of the prior and the diagnostic likelihood to be calibrated. However, tempering does not remove structural errors from a model. The reliability of the corrected belief therefore depends on both diagnostic calibration and degradation-model estimation.

6.3 Maintenance results

The maintenance results suggest that improved state estimation can support better replacement timing under the tested cost structure. This result should be interpreted cautiously. The cost advantage is obtained under the assumed values of C_P and C_C . In addition, CBR1 has three corrective replacements that are not observed under CBR0. Different cost ratios, risk preferences, or decision rules could lead to different conclusions.

Therefore, while the corrected belief improves hidden-state estimation in the case study, the maintenance benefit depends on how that belief is translated into operational decisions and on the economic context in which these decisions are made.

6.4 Relationship to existing research

These findings are consistent with previous research on bearing diagnostics and partially observed maintenance. Prior studies show that vibration-based bearing diagnostics can be uncertain because early faults are weak and diagnostic signatures can be masked by noise (Randall & Antoni, 2011; Taghiyarrenani & Berenji, 2022; Tandon & Choudhury, 1999).

The strong class dependence is also consistent with uncertainty-aware diagnostic research by Nemani et al. (2023), where classification reliability can vary across fault modes. The poor outer-race classification is consistent with Lee and Fecarotti (2026), who report a high false-negative rate for the outer-race fault mode. The present results extend that finding by showing that temporal degradation information can partly compensate for such diagnostic weaknesses, especially when diagnostic outputs are ambiguous.

6.5 Limitations

The study has three main limitations. First, the evaluation is simulation-based and uses a monotone, single-fault degradation structure. This is simpler than many physical degradation processes. Real systems may show interacting faults, variable operating conditions, non-monotone diagnostic signatures, or measurement effects that resemble recovery. These effects would require a richer state process.

Second, diagnostic uncertainty is represented by a confusion matrix. This makes the diagnostic uncertainty easy to integrate, but it discards information that may be present in the full output distribution. More complex models could provide more informative diagnostic evidence than sampled hard labels.

Third, the transition matrix is estimated from right-censored data. This is an intentional limitation, but it limits how much can be learned about internal state dwell times. More frequent inspection data, larger datasets, or datasets with better coverage across fault branches would improve the estimation of transition probabilities.

Overall, the proposed fusion procedure is most appropriate when degradation is sequential, diagnostic outputs are uncertain, and the transition dynamics can be estimated with sufficient reliability. Its main contribution is to embed class-dependent diagnostic uncertainty within a temporally consistent inference framework. Its main limitation is that the quality of the corrected belief depends on the quality of both inputs.

7 Conclusion

This study shows that a Markov–diagnostic fusion procedure can improve hidden-state estimation for condition-based maintenance when diagnostic information is uncertain and degradation dynamics are only indirectly observed.

The main finding is that the corrected-belief estimate improves state estimation relative to the diagnostic baseline in the ball-bearing case study. The improvement is especially relevant when the diagnostic model has difficulty distinguishing between degradation states and the degradation history provides additional information about which states are plausible. This supports the value of combining class-dependent diagnostic evidence with sequential degradation dynamics when the true component condition is only partially observed.

The maintenance evaluation shows that the corrected operating belief can also support replacement timing under the tested threshold policies and cost parameters. The corrected-belief threshold policy lowers the held-out cost rate mainly by allowing longer operating cycles. However, this result should be interpreted as case-specific. The cost benefit depends on the chosen decision rule, cost structure, and tolerance for corrective replacements.

The study also shows that the quality of the estimated transition matrix is important. The ball-defect branch demonstrates that the fusion procedure can introduce model-based bias when part of the estimated transition matrix is weakly informed. The practical value of the fusion procedure therefore depends on both diagnostic calibration and reliable estimation of the degradation dynamics.

Future research should evaluate the fusion procedure under richer degradation settings, including larger datasets, more frequent inspection points, branch-imbalanced fault data, and diagnostic models that provide calibrated probabilistic outputs rather than hard labels. Overall, the fusion procedure should not be viewed as a replacement for diagnostic modelling or transition estimation. Instead, it provides a way to embed uncertain diagnostic information within a temporally consistent inference framework that can support condition-based maintenance decisions.

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A EM algorithm

This appendix gives the expectation–maximization (EM) recursions used to estimate the transition matrix P from right-censored data for the two-severity Markov structure used in this study. Let $A(i)$ denote the set of admissible successor states from state i .

The data contain two types of observations. Let $\mathcal{R}_F$

denote the set of failed units and let $\mathcal{R}_C$ denote the set of censored units. For failed units $r \in \mathcal{R}_F$, the observed data consist of a failure mode $M_r \in M$ and a failure time $\tau_{F,r}$. For censored units $r \in \mathcal{R}_C$, the observed data consist of a censoring time $\tau_{C,r}$ and an observed transient endpoint state $S_{\tau_{C,r}}^{(r)} \in T$. Here, $S_t^{(r)}$ denotes the true degradation state of unit r at time t .

Define the observed failure-time and censoring-time cell sets as

$$\mathcal{T}_F = \{\tau_{F,r} : r \in \mathcal{R}_F\}, \quad \mathcal{T}_C = \{\tau_{C,r} : r \in \mathcal{R}_C\}.$$

So, $t \in \mathcal{T}_F$ denotes an observed failure-time cell, and $c \in \mathcal{T}_C$ denotes an observed censoring-time cell.

In the ball-bearing numerical case study, $M = \{O, I, B\}$ and

$$T = \{H, O_1, O_2, I_1, I_2, B_1, B_2\}.$$

For $m \in M$, $t \in \mathcal{T}_F$, the grouped failed-unit counts are

$$n_F(m, t) = \#\{r \in \mathcal{R}_F : M_r = m, \tau_{F,r} = t\},$$

and for $s \in T$, $c \in \mathcal{T}_C$ the grouped censored-unit counts are

$$n_C(s, c) = \#\{r \in \mathcal{R}_C : S_{\tau_{C,r}}^{(r)} = s, \tau_{C,r} = c\}.$$

Let Q_m denote the transient submatrix of P restricted to the states $\{H, m_1, m_2\}$. Let f_m denote the vector of transition probabilities from these states to F .

$$Q_m = [\{H, m_1, m_2\}, \{H, m_1, m_2\}],$$

$$f_m = [\{H, m_1, m_2\}, F].$$

For censored observations, define the full transient block

$$Q = P[T, T].$$

At EM iteration k , let $P^{(k)}$ denote the current transition matrix. The corresponding matrices and failure vector are denoted by $Q_m^{(k)}$, $Q^{(k)}$, and $f_m^{(k)}$.

A.1 Failed observations

For a failed mode-time cell (m, t) , with $m \in M$ and $t \in \mathcal{T}_F$, define the normalizing likelihood

$$Z_F^{(k)}(m, t) = e_H^\top \left(Q_m^{(k)} \right)^{t-1} f_m^{(k)}. \quad (\text{A.1.1})$$

This is the probability of reaching failure in mode m exactly at time t , starting from the healthy state.

Define the forward and continuation quantities

$$\lambda_u^{(m)} = e_H^\top \left(Q_m^{(k)} \right)^u, \quad u = 0, \dots, t-1, \quad (\text{A.1.2})$$

and

$$\rho_v^{(m)} = \left(Q_m^{(k)} \right)^{v-1} f_m^{(k)}, \quad v \geq 1. \quad (\text{A.1.3})$$

Here, $\lambda_u^{(m)}(i)$ is the probability of occupying state $i \in \{H, m_1, m_2\}$ after u transient steps, while $\rho_v^{(m)}(j)$ is the probability of reaching failure in exactly v further steps when currently in state $j \in \{H, m_1, m_2\}$.

For an admissible transient transition $i \rightarrow j$ within $\{H, m_1, m_2\}$, the expected transition count from failed observations in cell (m, t) is written as $\bar{N}_{ij}^{F, m, t}$, with

$$\bar{N}_{ij}^{F, m, t} = n_F(m, t) \sum_{u=0}^{t-2} \frac{\lambda_u^{(m)}(i) p_{ij}^{(k)} \rho_{t-u-1}^{(m)}(j)}{Z_F^{(k)}(m, t)}. \quad (\text{A.1.4})$$

The final transition into failure is handled separately. For an admissible transition $i \rightarrow F$,

$$\bar{N}_{iF}^{F, m, t} = n_F(m, t) \frac{\lambda_{t-1}^{(m)}(i) p_{iF}^{(k)}}{Z_F^{(k)}(m, t)}. \quad (\text{A.1.5})$$

Under the transition structure used in this paper, only $m_2 \rightarrow F$ is admissible within branch m . Therefore, each failed unit contributes one final transition into failure.

A.2 Censored observations

For a censored state-time cell (s, c) , with $s \in T$ and $c \in \mathcal{T}_C$, define the normalizing likelihood

$$Z_C^{(k)}(s, c) = e_H^\top \left(Q^{(k)} \right)^c e_s. \quad (\text{A.2.1})$$

This is the probability that a unit starting in the healthy state is observed in transient endpoint state s at censoring time c , without having reached failure before that time.

Define the forward quantities

$$\lambda_u = e_H^\top \left(Q^{(k)} \right)^u, \quad u = 0, \dots, c, \quad (\text{A.2.2})$$

and, for a fixed endpoint state $s \in T$, define

$$\rho_v^{(s)} = \left(Q^{(k)} \right)^v e_s, \quad v = 0, \dots, c. \quad (\text{A.2.3})$$

Here, $\lambda_u(i)$ is the probability of occupying transient state i after u steps, and $\rho_v^{(s)}(j)$ is the probability of reaching endpoint state s in exactly v further transient steps when currently in state j .

For an admissible transient transition $i \rightarrow j$ in T , the expected transition count from censored observations in cell (s, c) is written as $\bar{N}_{ij}^{C, s, c}$, with

$$\bar{N}_{ij}^{C, s, c} = n_C(s, c) \sum_{u=0}^{c-1} \frac{\lambda_u(i) p_{ij}^{(k)} \rho_{c-u-1}^{(s)}(j)}{Z_C^{(k)}(s, c)}. \quad (\text{A.2.4})$$

Censored observations do not contribute transitions into failure, since the unit is observed to be operating at the censoring time.

A.3 M-step

The total expected transition count is obtained by summing the failed and censored contributions over

all observed cells:

$$\bar{N}_{ij}^{(k)} = \sum_{m \in M} \sum_{t \in \mathcal{T}_F} \bar{N}_{ij}^{F, m, t} + \sum_{s \in T} \sum_{c \in \mathcal{T}_C} \bar{N}_{ij}^{C, s, c}. \quad (\text{A.3.1})$$

For transitions into failure, only the failed-observation terms contribute, because censored units are observed in transient states.

In the M-step, each admissible row of the transition matrix is updated by normalizing the expected transition counts:

$$p_{ij}^{(k+1)} = \frac{\bar{N}_{ij}^{(k)}}{\sum_{\ell \in A(i)} \bar{N}_{i\ell}^{(k)}}, \quad j \in A(i). \quad (\text{A.3.2})$$

All structurally inadmissible transitions remain fixed at zero. The failure state remains absorbing, so

$$p_{FF}^{(k+1)} = 1. \quad (\text{A.3.3})$$

If the denominator for a row is zero, the row is left unchanged from $P^{(k)}$ to avoid division by zero.

The EM algorithm is iterated until the change in the observed-data log-likelihood $\ell_{\text{obs}}(P)$ and the maximum entrywise change in P fall below the convergence tolerance, or until the maximum number of iterations is reached.

B Implementation details

This appendix reports implementation details that are required to reproduce the EM algorithm, the tuning of the state-fusion exponents, and the maintenance thresholds.

B.1 Transition-matrix EM implementation

For the reported mode-augmented right-censored fit, the EM algorithm is initialized with the symmetric feasible matrix below, with row and column order $(H, O_1, O_2, I_1, I_2, B_1, B_2, F)$:

$$P^{(0)} = \begin{bmatrix} 0.25 & 0.25 & 0 & 0.25 & 0 & 0.25 & 0 & 0 \\ 0 & 0.5 & 0.5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0 & 0 & 0 & 0.5 \\ 0 & 0 & 0 & 0.5 & 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.5 & 0 & 0 & 0.5 \\ 0 & 0 & 0 & 0 & 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.5 & 0.5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Structural zeros are enforced by updating only admissible transitions, and F is kept absorbing.

Failed observations are grouped by realized fault mode and failure time. All 50 failures are mode-labeled: 15 outer-race, 20 inner-race, and 15 ball-defect failures. Censored observations are grouped by endpoint state and censoring time. The data contain 50 censored endpoints at $t = 20$ and no survival-only censored rows.

At each E-step, expected counts are computed by deterministic forward recursions. Mode-labeled failures use the branch-local states $\{H, m_1, m_2\}$. Censored endpoint rows use the full transient block Q .

At each M-step, admissible transition probabilities are updated by row-normalizing the expected transition counts. If a row has zero expected source count, the previous row is retained.

Iteration stops when both $|\Delta\ell_{\text{obs}}|$ and the maximum entrywise matrix change are at most 2×10^{-6} , or when 2000 iterations are reached. The model converged after 908 iterations, with final observed-data log-likelihood -341.736879 . The EM uses double-precision forward probabilities, checks for nonpositive or nonfinite likelihood denominators, and retains previous rows to avoid division by zero. It does not use a likelihood floor or log-space forward recursion in the EM step.

B.2 State-fusion exponent tuning

The prior exponent α and the diagnostic-likelihood exponent β are selected using only the training trajectories. The case-study split is 700 training bearings and 300 held-out evaluation bearings. The candidate grid is

$$\alpha, \beta \in \{0, 0.2, 0.4, \dots, 2.0\}.$$

For each candidate pair, the fusion procedure is applied to the training trajectories, and the corrected-belief estimate is compared with the true simulated state over pre-failure operating observations. Failure observations are excluded from the tuning accuracy calculation.

The selected exponent pair maximizes training state-estimation accuracy. If several exponent pairs tie, the implementation prefers the pair closest to $(1, 1)$, then the pair with smaller $|\alpha - \beta|$, then smaller $\alpha + \beta$, then smaller α , then smaller β . The selected values are $\lambda = 1$ and $\beta = 1.8$, with training-split selection accuracy 0.7736. These values are fixed before held-out evaluation.

B.3 Maintenance-threshold tuning

Maintenance thresholds are selected using the training trajectories. The preventive replacement cost is $C_P = 10$, and the corrective replacement cost is $C_C = 50$. The threshold candidates are

$$\eta_O, \eta_I, \eta_B \in \{0, 0.1, 0.2, \dots, 1.0\}.$$

CBR0 and CBR1 thresholds are tuned separately. For CBR1, the prior exponent $\alpha = 1$ and diagnostic-likelihood exponent $\beta = 1.8$ are fixed before threshold tuning.

For each threshold triple, the policy is evaluated on the training trajectories, and the selected triple minimizes training cost rate,

$$\text{Cost rate} = \frac{\text{total maintenance cost}}{\text{total operating time}}.$$

If several triples tie, the implementation minimizes cost rate, then average cost, then prefers larger average cycle length, then larger thresholds in the order η_O, η_I, η_B .

The selected thresholds for the diagnostic-evidence threshold policy are fixed before the held-out test evaluation.

C Uncertainty and sensitivity analysis

This appendix reports additional uncertainty and sensitivity checks for the held-out evaluation to assess the stability of the main state-estimation and maintenance results. The primary uncertainty analysis uses trajectory-level bootstrap resampling of the held-out test set. These intervals represent conditional held-out performance uncertainty. They condition on the fitted transition matrix, the empirical confusion matrix, the selected state-fusion exponents, and the selected maintenance thresholds.

C.1 Trajectory-level bootstrap of held-out performance

Table 4 reports paired bootstrap differences between the corrected-belief results and the corresponding diagnostic baseline. Positive values indicate higher accuracy or longer cycle length for the corrected-belief results. For cost rate, negative values indicate lower cost.

Table 4: Trajectory-level bootstrap differences on the held-out test set.

Metric	Difference	95% CI
Accuracy	0.162	[0.132, 0.193]
Balanced accuracy	0.047	[0.030, 0.067]
Severity-2 exact accuracy	0.508	[0.440, 0.571]
Cost rate	-0.048	[-0.079, -0.014]
Cycle length	1.557	[1.267, 1.867]
Corrective replacements	3.000	[0.000, 7.000]

The bootstrap intervals support the main result that the corrected-belief estimate improves overall accuracy. However, the corrected-belief threshold policy introduces corrective replacements, with an interval between 0 and 7 replacements. This supports the main interpretation that the maintenance improvement is economic, but not risk-free.

C.2 Diagnostic confusion-matrix uncertainty

The empirical CNN confusion matrix is estimated from 15 held-out test samples per class. Table 5 reports class-wise recall uncertainty using binomial Wilson intervals. The intervals are wide because of the small number of test samples per class.

The diagnostic recall uncertainty confirms that the 1D-CNN diagnostic model performs unevenly across states. The outer-race severity states have low estimated recall, with O_2 not correctly identified in the held-out CNN test set. This supports the use of temporal degradation information, but also shows that the

Table 5: Class-wise CNN recall uncertainty.

Class	Recall	95% CI
H	1.000	[0.796, 1.000]
O_1	0.200	[0.070, 0.452]
O_2	0.000	[0.000, 0.204]
I_1	0.933	[0.702, 0.988]
I_2	1.000	[0.796, 1.000]
B_1	1.000	[0.796, 1.000]
B_2	0.933	[0.702, 0.988]

confusion-matrix observation model is estimated from a limited test sample.

C.3 Sensitivity checks

Tables 6 and 7 summarize sensitivity checks for the likelihood floor, state-fusion exponents, and corrective-to-preventive cost ratio. The cost-ratio comparison keeps the replacement decisions fixed and recomputes the resulting cost rate under alternative values of C_C/C_P ; thresholds are not retuned for each cost ratio.

Table 6: State-estimation sensitivity checks for selected modelling assumptions.

Sensitivity	Setting	Accuracy	Bal. acc.	Sev.-2 acc.
Likelihood floor	$\epsilon_G = 10^{-6}$	0.790	0.772	0.808
Likelihood floor	$\epsilon_G = 10^{-5}$	0.790	0.772	0.808
Likelihood floor	$\epsilon_G = 10^{-4}$	0.790	0.772	0.808
Likelihood floor	$\epsilon_G = 10^{-3}$	0.790	0.772	0.807
Fusion exponents	$(\alpha, \beta) = (0.8, 1.8)$	0.770	0.761	0.857
Fusion exponents	$(\alpha, \beta) = (1.0, 1.6)$	0.791	0.773	0.822
Fusion exponents	$(\alpha, \beta) = (1.0, 1.8)$	0.790	0.772	0.808
Fusion exponents	$(\alpha, \beta) = (1.0, 2.0)$	0.793	0.774	0.805
Fusion exponents	$(\alpha, \beta) = (1.2, 1.8)$	0.780	0.763	0.685

Table 7: Cost-ratio sensitivity checks for the maintenance policies.

C_C/C_P	Policy	Cost rate	Corr.
2	Diagnostic-evidence	0.730	0
2	Corrected-belief	0.662	3
5	Diagnostic-evidence	0.730	0
5	Corrected-belief	0.682	3
10	Diagnostic-evidence	0.730	0
10	Corrected-belief	0.714	3

The likelihood-floor sensitivity indicates that the state-estimation result is essentially unchanged across the tested ϵ_G values. The state-fusion exponent sensitivity shows that performance is stable around the selected value $(\alpha, \beta) = (1.0, 1.8)$, although increasing the prior exponent to $\alpha = 1.2$ reduces severity-2 performance. The cost-ratio sensitivity shows that the corrected-belief threshold policy retains a lower cost rate for the tested cost ratios, but the difference becomes smaller as corrective replacement becomes more expensive.